Stochastic optimization
Stochastic dual dynamic programming

Fabian Bastin
bastin@iro.umontreal.ca
Université de Montréal – CIRRELT – IVADO – Fin-ML

Context

Introduced by Pereira and Pinto (1991), first designed to solve hydro-scheduling problems.

- Stochastic dynamic problem with finite horizon.
- Continuous, finite dimensional state and control.
- Convex cost, additive over stages – we focus here on linear costs.
- Discrete, stagewise independent noises.

SDDP builds on the nested L-shaped method of the Multistage slides, recalled in the next three slides.

Julia implementations: the `SDDP.jl` package (examples: `code/SDDP_hydro.ipynb`), and the step-by-step notebook `code/SDDP_from_scratch.ipynb`.

These slides are based on Anthony Papavasiliou's material, `https://ap-rg.eu/wp-content/uploads/2020/05/SDDP.pdf`.

Problem setting

We consider the multistage stochastic linear program of the Multistage slides, over the horizon $t = 1, \ldots, H$ (we write H, and not T, since T denotes the technology matrix), with the constraints $W_1 x_1 = h_1$ and $W_t x_t = h_t(\xi_t) - T_{t-1}(\xi_t) x_{t-1}$, $t = 2, \ldots, H$. In dynamic programming form, with $\mathcal{Q}_{H+1} \equiv 0$,

$$V_t(x_{t-1}, \xi_{[t]}) = \min_{x_t \geq 0} \left\{ c_t^T x_t + \mathcal{Q}_{t+1}(x_t, \xi_{[t]}) \ : \ W_t x_t = h_t(\xi_t) - T_{t-1}(\xi_t) x_{t-1} \right\},$$

$$\mathcal{Q}_{t+1}(x_t, \xi_{[t]}) = \mathbb{E}\left[V_{t+1}(x_t, \xi_{[t+1]}) \mid \xi_{[t]} \right].$$

The expected cost-to-go is convex in x_t, and piecewise linear on its domain: SDDP builds it from below, one supporting hyperplane at a time (background notes). Under stagewise independence, the conditioning disappears and a single $\mathcal{Q}_{t+1}(x_t)$ serves the whole stage.

Recap: nested L-shaped method

At node $k = 1, \ldots, N_t$ of stage t, nested decomposition solves the problem $NLSD(t, k)$

$$
\begin{aligned}
\min_{x_{t,k} \geq 0,\, \theta_{t,k}} \; & c_{t,k}^T x_{t,k} + \theta_{t,k} \\
\text{s.t.} \; & W_t x_{t,k} = h_{t,k} - T_{t-1,k}\, \hat{x}_{t-1,a(k)}, \quad (\pi_{t,k}) \\
& D_{t,k,\ell}\, x_{t,k} \geq d_{t,k,\ell}, \;\; \ell = 1, \ldots, r_{t,k}, \quad (\rho_{t,k,\ell}) \quad \text{feasibility cuts,} \\
& E_{t,k,\ell}\, x_{t,k} + \theta_{t,k} \geq e_{t,k,\ell}, \;\; \ell = 1, \ldots, s_{t,k}, \quad (\sigma_{t,k,\ell}) \quad \text{optimality cuts.}
\end{aligned}
$$

- $NLSD$: nested L-shaped decomposition subproblem; N_t: number of nodes at stage t; $a(k)$: parent of node k; $\theta_{t,k}$: approximation of the expected cost-to-go, bounded below until its first cut (by 0 if the costs are nonnegative), $\theta_{H,k} \equiv 0$.
- Forward pass: the trial decisions go down the tree. Backward pass: for $t = H, \ldots, 2$, the nodes of stage t send cuts to their parents. The first-stage value is a lower bound, the expected cost of the policy an upper bound. A feasibility cut comes from a certificate of infeasibility of a child; none is needed with relatively complete recourse.

Recap: optimality cuts

For a node j at stage $t-1$, solve $NLSD(t,k)$ at $\hat{x}_{t-1,j}$ for each of its children $k \in \mathcal{D}_t(j)$, with optimal value $v_{t,k}$, and compute

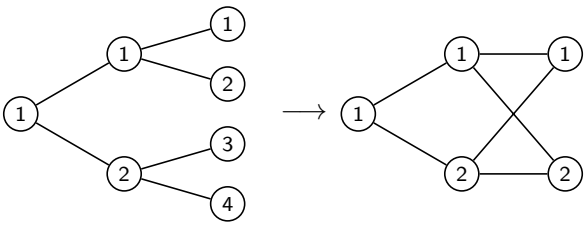
$$E = \sum\nolimits_{k \in \mathcal{D}_t(j)} p(k, t \,|\, j, t-1)\, \pi_{t,k}^T T_{t-1,k},$$

$$e = \sum\nolimits_{k \in \mathcal{D}_t(j)} p(k, t \,|\, j, t-1) \left(v_{t,k} + \pi_{t,k}^T T_{t-1,k}\, \hat{x}_{t-1,j} \right),$$

with $p(k, t \,|\, j, t-1)$ the probability of moving from j to k, then add the cut $E x_{t-1,j} + \theta_{t-1,j} \geq e$ to $NLSD(t-1, j)$.

- This is the cut of the Multistage slides: $e - E\hat{x}_{t-1,j}$ is the average of the $v_{t,k}$. By linear programming duality, $v_{t,k} + \pi_{t,k}^T T_{t-1,k} \hat{x}_{t-1,j} = \pi_{t,k}^T h_{t,k} + \sum_\ell \rho_{t,k,\ell}\, d_{t,k,\ell} + \sum_\ell \sigma_{t,k,\ell}\, e_{t,k,\ell}$ as long as the lower bound on $\theta_{t,k}$ is inactive.
- It supports, at $\hat{x}_{t-1,j}$, the average of the children's current approximations, which lie below V_t: it is valid. It touches $\mathcal{Q}_t$ at $\hat{x}_{t-1,j}$ only if $v_{t,k} = V_t(\hat{x}_{t-1,j}, \xi_{[t]})$ for every child, as at stage H, where $\theta_{H,k} \equiv 0$.

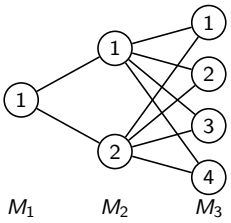
Recombined scenario tree



- When can we recombine nodes?
- When can we assign the same value function $\mathcal{Q}_{t+1}(x)$ to each node k of stage t?

Nested decomposition is non-scalable

Assume H stages, M_t discrete outcomes at stage t ($M_1 = 1$), and relatively complete recourse (no feasibility cut needed). The tree then has $N_t = \prod_{j=1}^{t} M_j$ nodes at stage t.



- Forward pass: $M_1 + M_1 \times M_2 + \ldots = \sum_{t=1}^{H} \prod_{j=1}^{t} M_j$ linear programs;
- Backward pass: $\sum_{t=1}^{H-1} \prod_{j=1}^{t} M_j$ linear programs, the nodes with children solved again with their new cuts.

The effort grows exponentially with the horizon.

Nested decomposition vs extensive form

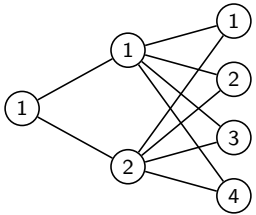
The alternative to nested decomposition is the extensive form.

- The extensive form will not even load in memory.
- Nested decomposition will load in memory, but will not terminate (for large problems).

Nested decomposition lays the foundation for SDDP.

As the method name suggests, we are in the context of dynamic programming.

Enumerating versus simulating



- Enumeration: $\{(1,1),(1,2),(1,3),(1,4),(2,1),(2,2),(2,3),(2,4)\}$;
- Simulation (with 3 samples): $\{(1,3),(2,1),(1,4)\}$.

Making nested decomposition scalable

Solution for the forward pass:

- in the forward pass, we simulate instead of enumerating;
- this results in a probabilistic upper bound and termination criterion.

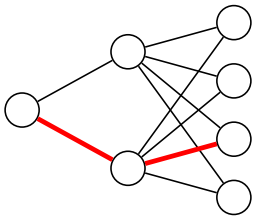
Solution for the backward pass:

- in the backward pass, we share cuts among nodes of the same time period;
- this requires an assumption of stagewise independence.

Stagewise independence: the random vector ξ_{t+1} is independent of $\xi_{[t]} = (\xi_1, \ldots, \xi_t)$ for $t = 1, \ldots, H-1$ (see for instance A. Shapiro, "Analysis of stochastic dual dynamic programming method", European Journal of Operational Research, 209(1), 2011, pp. 63–72). Here, ξ_1 has a unique realization.

Implications for the forward pass

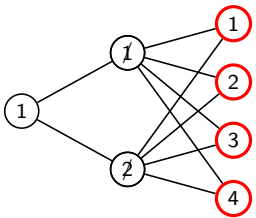
At each forward pass we solve $H - 1$ subproblems along a single sampled scenario.



For K Monte Carlo simulations, we solve $1 + K \times (H - 1)$ linear programs, whatever the size of the scenario tree.

Implications for the backward pass

Stagewise independence implies the same value function for all nodes of stage $t \Longrightarrow$ cut sharing.



For a given trial sequence $\hat{x}_t$, $t = 1, \ldots, H-1$, we solve $\sum_{t=2}^{H} M_t$ linear programs, and for K trial sequences, $K \sum_{t=2}^{H} M_t$ linear programs: the effort is now linear in the horizon.

Stagewise independence is not necessary

We can use dual multipliers in stage $t+1$ for cuts in stage t even without stagewise independence.

However, each node in stage t has a different value function, so

- more memory consumption;
- more optimality cuts needed because we are approximating more value functions.

With stagewise independence, we can

- get rid of the scenario tree;
- work with a continuous distribution of ξ_t, replaced by a sample of it in the backward pass (Shapiro, 2011).

An intermediate solution consists in adding ξ_t to the state vector, for instance when the noise follows an autoregressive model.

SDDP forward pass

- Solve $NLSD(1,1)$. Let $x_{1,1}$ be the optimal solution. Initialize $\hat{x}_{1,i} = x_{1,1}$ for $i = 1, \ldots, K$.
- For $t = 2, \ldots, H$, $i = 1, \ldots, K$,
  - sample a realization $\xi_{t,i}$ among $\xi_{t,k}$, $k = 1, \ldots, M_t$, with probabilities $p_{t,k}$, which determines $(c_{t,i}, h_{t,i}, T_{t-1,i})$;
  - solve $NLSD(t,i)$ with the trial decision $\hat{x}_{t-1,i}$;
  - store the optimal solution as $\hat{x}_{t,i}$, and the stage cost $c_{t,i}^T \hat{x}_{t,i}$.

With shared cuts, $NLSD(t,k)$ depends on the node only through its realization: k now indexes the M_t realizations of ξ_t, and $NLSD(t,i)$ is the problem of the realization drawn in scenario i. The stage costs exclude $\theta_{t,i}$: otherwise the future would be counted twice.

SDDP backward pass

- For $t = H, H-1, \ldots, 2$, for $i = 1, \ldots, K$,
  - solve $NLSD(t, k)$ with the trial decision $\hat{x}_{t-1,i}$, for $k = 1, \ldots, M_t$, with optimal value $v_{t,k,i}$ and multipliers $\pi_{t,k,i}$;
  - compute

$$E_{t-1,i} = \sum_{k=1}^{M_t} p_{t,k}\, \pi_{t,k,i}^T T_{t-1,k}, \qquad e_{t-1,i} = \sum_{k=1}^{M_t} p_{t,k} \left(v_{t,k,i} + \pi_{t,k,i}^T T_{t-1,k}\, \hat{x}_{t-1,i} \right);$$

  - add the optimality cut $E_{t-1,i}\, x_{t-1} + \theta_{t-1} \geq e_{t-1,i}$ to $NLSD(t-1, k)$, $k = 1, \ldots, M_{t-1}$.

The same cut is added to every node of stage $t-1$: this is cut sharing. Under stagewise independence, $p_{t,k}$ is the probability $p(k, t \,|\, j, t-1)$ of the recap, whatever the node j.

Probabilistic upper bound

Suppose we draw a scenario i, that is a realization of (ξ_t), $t = 2, \ldots, H$, and solve $NLSD(t, i)$ for $t = 1, \ldots, H$ (noting that for $t = 1$, the problem is deterministic).

- This gives $x_{t,i}$, $t = 1, \ldots, H$, with a cost $z_i = \sum_{t=1}^{H} c_{t,i}^T x_{t,i}$.
- Each $(x_{t,i},\ t = 1, \ldots, H)$ is feasible but not necessarily optimal, so the expected cost $\mu = \mathbb{E}[z_i]$ of the current policy is an upper bound on the optimal value.
- Consider K independent scenarios, and the mean cost $\overline{z}_K = \frac{1}{K} \sum_{i=1}^{K} z_i$. By the Central Limit Theorem,

$$\sqrt{K}\,(\overline{z}_K - \mu) \xrightarrow{\mathcal{D}} N(0, \sigma^2), \quad \text{i.e., } \overline{z}_K \approx N\left(\mu, \frac{\sigma^2}{K}\right).$$

- $\overline{z}_K$ only estimates that upper bound: a realization of $\overline{z}_K$ may well fall below the optimal value.

Bounds and termination criterion

After solving $NLSD(1,1)$ in a forward pass we can compute a lower bound z_{LB} as the objective function value of $NLSD(1,1)$: the cuts are an outer approximation of the expected cost-to-go, hence $NLSD(1,1)$ is a relaxation of the original problem.

Terminate if $z_{LB} \in (\overline{z}_K - \Phi^{-1}(1-\alpha/2)\hat{s}/\sqrt{K}, \overline{z}_K + \Phi^{-1}(1-\alpha/2)\hat{s}/\sqrt{K})$, where $\Phi(\cdot)$ is the cumulative distribution of a Normal(0,1), and $1-\alpha$ is the confidence level set by the user. Typically, $\alpha = 0.05$. Since z_{LB} is a lower bound, only the left inequality really matters.

How large should K be? If the half-width $\Phi^{-1}(1-\alpha/2)\hat{s}/\sqrt{K}$ is about $0.01\,\overline{z}_K$, the test only stops when $\overline{z}_K - z_{LB} < 0.01\,\overline{z}_K$. Using the empirical standard deviation $\hat{s} = \sqrt{\frac{1}{K-1}\sum_{i=1}^{K}(z_i - \overline{z}_K)^2}$, this leads to

$$K = \left(\frac{\Phi^{-1}(1-\alpha/2)\hat{s}}{0.01\,\overline{z}_K} \right)^2 .$$

The criterion is optimistic: with a small K the confidence interval is wide, and the lower bound enters it long before the policy is optimal.

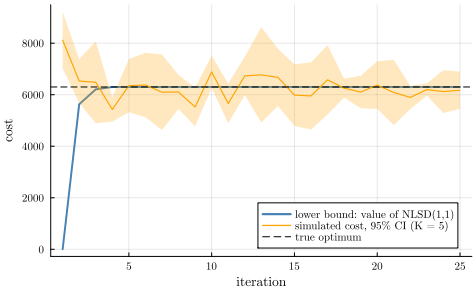
SDDP algorithm

Set K, α, the maximum number of iterations; start with no cut, each θ_t bounded below.
repeat
  Forward pass: solve $NLSD(1,1)$, set z_{LB} to its optimal value;
    simulate K scenarios, collect the trial solutions $\hat{x}_{t,i}$ and the costs z_i;
    compute $\overline{z}_K$ and $\hat{s}$.
  if $z_{LB} \geq \overline{z}_K - \Phi^{-1}(1-\alpha/2)\hat{s}/\sqrt{K}$ **then**
    stop: the gap is not statistically significant.
  end if
  Backward pass: for $t = H, \ldots, 2$ and $i = 1, \ldots, K$, solve the M_t
    subproblems at $\hat{x}_{t-1,i}$ and add the resulting cut to stage $t-1$.
until the iteration limit is reached

The lower bound is deterministic and non-decreasing; the upper bound is statistical.

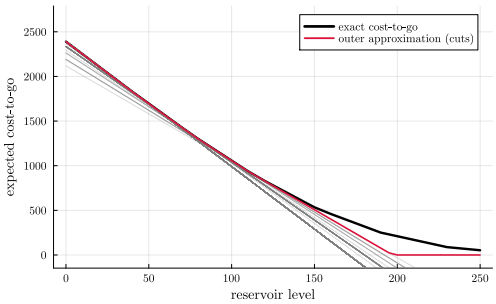
Numerical illustration

Hydro-thermal scheduling, $H = 6$ stages, $M_t = 4$ inflow outcomes (1024 scenarios), $K = 5$ forward passes per iteration.



The lower bound reaches the optimal value of the extensive form after a few iterations, while the simulated cost keeps oscillating around it.

Numerical illustration: the cuts



Exact expected cost-to-go $\mathcal{Q}_5$ and the cuts of stage 4. Each cut lies below $\mathcal{Q}_5$, and the approximation is tight only where the forward passes went: the value function is never learned everywhere. Computed with `code/SDDP_from_scratch.ipynb`.

References

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- A. Shapiro, "Analysis of stochastic dual dynamic programming method", *European Journal of Operational Research* 209(1), 63–72, 2011.
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- A. Papavasiliou, lecture notes on SDDP, `https://ap-rg.eu/wp-content/uploads/2020/05/SDDP.pdf`.